Charter Brokerage Platform – Requirements Brief

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1. Purpose and scope

We are building a web platform that lets a private-jet charter brokerage run its whole client-facing business: agents submit client requests, the admin team sources aircraft from external marketplaces (Avinode and others), prices and sends quotes, and clients accept, sign, provide passenger data, pay and receive flight briefs.

The platform does not replace Avinode. Sourcing (searching aircraft and sending RFQs to operators) happens in Avinode by hand. The platform records the results and handles everything before and after.

Commercial constraints

- One-time development fee. Recurring running costs must stay low and predictable (hosting, domain, email, storage, payment fees). The developer provides a monthly run-cost estimate with the proposal; proposed target: under €100/month excluding payment-provider fees (to be confirmed).
- The client owns all source code, data, domains and accounts (hosting, payment provider, email). Full IP assignment in writing.
- Mainstream, open-source, well-documented stack so any competent developer can take over. No proprietary lock-in.
- Deliverables include a repository, deployment guide, handover document and a short admin user guide.

Language: English only (interface, documents, emails). Multi-currency, EUR as default.

2. Business model and legal role

- The company is the **contracting party**. It holds a contract with each client and a separate contract with each operator.
- The **client never sees the operator's name** at quote stage. See requirement OPR-6 for when identity is disclosed.
- Operators are **qualified by the company** against a defined standard (Section 8) before they can be used.

- Agents and partner brokers introduce and manage clients. They earn commission. They do not contract with operators.

> **Legal flag for the client to confirm with a lawyer, not a developer decision:** EU rules on informing passengers of the identity of the operating air carrier (Regulation (EC) 2111/2005, Art. 11) may require disclosure at booking or at the latest before departure. The system must therefore make the disclosure point **configurable** (default: disclosed in the confirmed flight brief). Also confirm contracting-carrier liability, VAT treatment and invoicing entity.

3. Users and roles

Role	Description	Key permissions
Super Admin	Company owner	Everything, including settings, users, margins, financial reports
Admin / Sales Desk	Staff handling requests	Process requests, source, price, send quotes, manage clients and agents, issue invoices
Finance	Staff handling money	Invoices, payments, credit notes, commission statements, reports
Operations	Trip support staff	Passenger data, flight briefs, trip checklists, handlers
Agent	External broker or partner	Submit requests, see own requests, own clients and quotes sent to them, own commission
Client (passenger/booker)	End customer	Access via secure link or account: view and accept quote, sign, enter passenger data, pay, view brief
Operator	Aircraft operator	**No login in Phase 1.** Records are kept by admins. Optional operator portal later

Data isolation: an agent sees only their own requests, clients and quotes. An agent can never see operator names, operator prices, or the company margin. Admin roles are configurable so permissions can be adjusted without code changes.

4. End-to-end workflow

1. **Request:** an agent (or admin on behalf of a client) submits a trip request.
2. **Queue:** the request appears in the admin queue and the desk is notified.
3. **Sourcing:** an admin searches Avinode manually and logs each operator response in the platform (operator, aircraft, price, validity, notes, attachments).
4. **Pricing:** the admin selects one or more options, the system applies margin rules and agent commission, and produces the client-facing quote.
5. **Delivery:** the quote is published to the agent's account (and optionally directly to the client). Agent and client are notified.
6. **Acceptance:** the client accepts through the portal and signs the terms electronically.
7. **Invoicing and payment:** the system issues the invoice and payment options. Payment is recorded (automatically for card, manually or via reconciliation for bank transfer).
8. **Passenger data:** the client or agent completes the passenger details through a secure form.
9. **Operator booking:** the admin confirms the booking with the operator (outside the system) and records the confirmation, aircraft registration and crew details if available.
10. **Flight brief:** the system generates the brief, and the client and agent can view or download it. Any change creates a new version and sends a notification.
11. **Post-flight:** trip closed, feedback recorded, operator scored, commission calculated.

5. Functional requirements

Priority: **M** = must (Phase 1), **S** = should (Phase 2), **C** = could (Phase 3).

5.1 Accounts, access and security

ID	Requirement	Pri
USR-1	Email and password login for staff and agents; two-factor authentication (TOTP) mandatory for staff, optional for agents	M
USR-2	Role-based access control as in Section 3; admin can create, deactivate and re-role users	M
USR-3	Agent onboarding: admin invites agent by email, agent completes profile (company, contact, tax ID, bank details for commission) and accepts the agent agreement	M

ID	Requirement	Pri
USR-4	Clients access via secure, expiring magic links (no password required) to a specific quote or trip; optional client account for repeat clients	M
USR-5	Password reset, session timeout, login-attempt throttling	M
USR-6	Full audit log: who changed what and when, for requests, quotes, prices, invoices, payments, user permissions. Not editable or deletable by users	M
USR-7	Agent white-label options: agent logo and colours on client-facing quotes	C

5.2 Agent portal

ID	Requirement	Pri
AGT-1	Request form: one-way, return and multi-leg itineraries; airports searchable by ICAO, IATA, city or name; date, time (flexible or fixed), passenger count, luggage, pets, special needs, aircraft category preference, budget indication, free-text notes, attachments	M
AGT-2	Save drafts; duplicate a previous request; submit	M
AGT-3	Client directory: create and edit own clients, companies and frequent passengers	M
AGT-4	Request list with status (Submitted, In sourcing, Quoted, Accepted, Paid, Confirmed, Flown, Cancelled, Expired) and filters	M
AGT-5	View quotes published to them; compare options side by side; download PDF; share a client link	M
AGT-6	Messaging per request (agent ⇄ admin desk), with email notification and file attachments	M
AGT-7	See flight brief, invoice and payment status of their own trips	M
AGT-8	Own commission overview: pending, due, paid, with downloadable statements	S

ID	Requirement	Pri
AGT-9	Agent-added markup on the client price, if the business decides to allow it (open decision D-1)	S
AGT-10	Mobile-friendly layout, installable as a progressive web app	M

5.3 Admin console

ID	Requirement	Pri
ADM-1	Request queue with priority, age, assignee, status and SLA timer; kanban and list views	M
ADM-2	Assign requests to staff; internal notes and tasks; internal chat separate from agent chat	M
ADM-3	**Sourcing log** per request: add any number of operator responses (operator, aircraft type and registration if known, base airport, price, currency, taxes and fees included or excluded, valid-until, notes, attachment such as PDF or email)	M
ADM-4	One-click button to copy request details formatted for pasting into Avinode, and a field to store the Avinode trip ID and link	M
ADM-5	Create a quote from selected sourcing options; multiple options per quote	M
ADM-6	Client and agent management with notes, history and documents	M
ADM-7	Dashboard: open requests, quotes awaiting response, expiring quotes, payments due, upcoming flights, monthly margin	M
ADM-8	Search across requests, clients, trips, invoices and airports	M
ADM-9	AI assistant to read a pasted operator email or quote and pre-fill the sourcing form for review (never auto-saves without admin confirmation)	S

ID	Requirement	Pri
ADM-10	Price benchmarking from stored quotes: typical price per route and aircraft category; warning if a new quote is far above or below history	S
ADM-11	Empty-leg watchlist: record empty legs found while sourcing and match them to open requests	C

5.4 Quote engine

ID	Requirement	Pri
QTE-1	Quote built from one or more sourcing options. Components: operator price, margin, handling and landing fees, catering, ground transport, de-icing, taxes, other extras. Every line item flagged as taxable or not	M
QTE-2	Margin rules: percentage or fixed amount, by default, by client, by agent, by route type or by aircraft category; manual override per quote with reason	M
QTE-3	Agent commission rule per agent (percentage of margin or fixed); shown only to admin and to that agent	M
QTE-4	Multi-currency with stored exchange rate at quote time; display currency chosen per quote	M
QTE-5	Validity period; automatic expiry and warning notifications to admin and agent	M
QTE-6	Versioning: any change creates a new version; previous versions kept and viewable by admins; client sees only the latest	M
QTE-7	Client-facing quote output as a web page and branded PDF: itinerary, aircraft category and type, seats, representative photos, price breakdown (as configured), terms, validity, cancellation policy, payment schedule. **Operator name is hidden**	M
QTE-8	Estimated block time and distance per leg (great-circle distance and aircraft cruise speed, with overrideable estimate)	M

ID	Requirement	Pri
QTE-9	Payment schedule per quote: full payment, or deposit plus balance with due dates	M
QTE-10	Estimated CO_2 per leg with optional offset line item	S
QTE-11	Quote templates, e.g. standard terms by region or client type	S
QTE-12	Quote tracking: sent, viewed (with timestamp), accepted, declined	M

5.4a Client portal

ID	Requirement	Pri
CLI-1	Secure link access (QR or short link) showing quote options, itinerary and price	M
CLI-2	Accept a quote option; decline with optional reason; request changes (creates a message to admin and agent)	M
CLI-3	**E-signature of terms and contract** built in: signer typed name and tick, timestamp, IP address, device, document hash, signed PDF stored and downloadable. Optional integration with a paid provider later	M
CLI-4	**Passenger data form**: per passenger full name as in passport, date of birth, nationality, passport number, issue and expiry dates, issuing country, contact details, dietary and special requests. Client or agent can complete it; save-and-return supported	M
CLI-5	Passport upload (optional); client can mark a passenger as complete; expiry validation (passport must be valid beyond the trip date plus a configurable period)	M
CLI-6	Payment page: invoice summary, card payment link, bank transfer details with unique reference; shows status	M

ID	Requirement	Pri
CLI-7	Flight brief access with version history; download PDF; "acknowledged" button	M
CLI-8	Trip changes: client can request changes or cancel, routed to admin	S
CLI-9	Post-flight feedback form	S

5.5 Documents

ID	Requirement	Pri
DOC-1	Template-driven PDF generation with company branding: quote, contract and terms, invoice, credit note, flight brief, passenger manifest, payment receipt	M
DOC-2	Templates editable by Super Admin (text blocks, logo, colours, footer, terms) without code changes	M
DOC-3	Invoices: sequential, gap-free numbering; compliant with Austrian and EU invoicing rules (seller and buyer details, VAT ID, date, description, VAT treatment); credit notes; configurable VAT treatment per line (confirmed by tax adviser)	M
DOC-4	**Flight brief** auto-assembled from trip data: itinerary with local and UTC times, aircraft type and registration (once confirmed), crew details if provided, passenger list, handler and FBO details per airport (contacts, location, hours, procedures), ground transport, catering, baggage allowance, customs and immigration notes, emergency contact, 24-hour operations number	M
DOC-5	Manifest export for operators and handlers (passenger data in required format), sent securely, with a log of who it was sent to	M
DOC-6	Every document stored per trip with version history	M
DOC-7	Send documents by email from the platform, with sent-log and open tracking	M

5.6 Payments and finance

ID	Requirement	Pri
PAY-1	Payment links via a mainstream provider (Stripe or Mollie, developer to recommend with fee comparison). Card, SEPA and others as supported	M
PAY-2	Bank transfer as a first-class method: details and unique payment reference on the invoice; Finance marks payments as received with date and amount; partial payments supported	M
PAY-3	Provider webhook updates payment status automatically; failed and refunded payments handled	M
PAY-4	Deposit and balance flow with automatic reminders before and after due dates	M
PAY-5	Card payments above a configurable threshold are flagged for Finance (chargeback risk); setting to restrict card payment to deposits only	S
PAY-6	Operator payables: record operator cost, operator invoice, payment due and paid; compare against quoted cost	S
PAY-7	Margin report per trip, per agent, per period; commission statements	S
PAY-8	Export of invoices and payments to CSV and to the accountant's format	S

5.7 Reference data

ID	Requirement	Pri
REF-1	**Airports:** import an open dataset such as OurAirports (verify current licence for commercial use). Fields: ICAO, IATA, name, city, country, latitude and longitude, elevation, time zone, runways (length, surface), frequencies, type. Admin can add or edit and mark private or custom fields; updates re-importable without losing edits	M

ID	Requirement	Pri
REF-2	**FBO / handler directory** linked to airports: handler name, address, phone, email, hours, services (customs, catering, hangar, crew car), PPR or slot requirements, fees notes, internal rating and notes, last-verified date. Starts small (airports actually used) and grows with use	M
REF-3	**Aircraft types:** manufacturer, model, category (turboprop, very light, light, midsize, super-midsize, large, ultra-long-range, airliner/VIP, helicopter), typical seats, range, cruise speed, cabin dimensions, baggage volume, runway needs, photo gallery (exterior, interior), notes	M
REF-4	**Specific aircraft (tails)**, optional: registration, year, interior refit, operator link, photos. Used internally and for flight brief	S
REF-5	Photo management: upload, tag, crop, order, mark licence source; photos are only shown if the licence permits. No scraping of third-party images	M
REF-6	Aircraft filter: "light jet, 6 seats, range over 1,500 nm" to support matching	M
REF-7	Country notes: visa, customs, permit requirements, overflight rules; free text per country	S

5.8 Operator management

ID	Requirement	Pri
OPR-1	Operator records: legal name, trading name, AOC number and issuing authority, base, contacts (sales, 24/7 ops), fleet, marketplace profile links	M
OPR-2	Qualification checklist per operator (Section 8) with document uploads, validity and expiry dates	M
OPR-3	Status: Pending, Approved, Approved with conditions, Suspended, Rejected. **Only Approved operators selectable in a quote**	M

ID	Requirement	Pri
OPR-4	Expiry alerts for AOC, insurance, audit rating and other time-limited documents (60 and 30 days before)	M
OPR-5	Operator contract (framework agreement) stored per operator with signed date	M
OPR-6	Operator identity visibility setting: hidden in quote; disclosure point configurable (default: shown in the confirmed flight brief). See legal flag in Section 2	M
OPR-7	Operator scorecard: response time, price competitiveness, reliability, safety incidents, client feedback; computed from trip data	S
OPR-8	Operator portal (upload documents, view confirmed trips)	C

5.9 Notifications and communication

ID	Requirement	Pri
NOT-1	Email notifications for: new request, quote ready, quote expiring, accepted, payment received, payment overdue, passenger data missing, brief published or updated, operator document expiring	M
NOT-2	Notification preferences per user	M
NOT-3	Email templates editable by Super Admin	M
NOT-4	In-app notification centre	S
NOT-5	WhatsApp or SMS notifications via a provider (usage-based cost, optional)	C
NOT-6	Pre-flight checklist reminders to Operations (handling, catering, transport, passenger data, permits)	S

6. Key data model (indicative)

The developer refines this. Principles: one **Trip request** has many **Legs**; one request has many **Sourcing options**; a **Quote** is built from options and has many **Versions**; an accepted quote creates a **Booking**, which owns **Passengers**, **Invoices**, **Payments**, **Documents** and **Flight brief versions**.

- **User, Role, Agent, Client, Company, Passenger**
- **Request, Leg** (origin, destination, date, time, pax)
- **SourcingOption** (operator, aircraft type, price, currency, validity, attachments, Avinode trip ID)
- **Quote, QuoteVersion, QuoteLineItem, MarginRule, CommissionRule**
- **Booking, Invoice, InvoiceLine, Payment, CreditNote**
- **Airport, Handler (FBO), AircraftType, Aircraft, Photo**
- **Operator, OperatorDocument, OperatorScore**
- **Document, Template, Notification, AuditLog, Message**

Cost and margin fields are visible to admin roles only and are never exposed through any agent or client endpoint.

7. Non-functional requirements

Security and privacy

- Hosted in the EU. HTTPS only. Encryption at rest for passenger and passport data and uploaded documents.
- GDPR: data processing records, defined retention (default: passenger identity data auto-deleted a configurable number of days after the flight; invoices kept as legally required), data export and deletion on request, minimal data collection.
- Passport data is visible only to roles that need it; every view of passport data is logged.
- OWASP Top 10 protections; rate limiting; secrets outside the repository; dependency updates.
- Role checks enforced on the server side for every endpoint (not just hidden in the interface).

Reliability and operations

- Automated daily backups with a tested restore procedure; at least 30 days of retention.
- Staging and production environments; deployment by script or pipeline; basic error monitoring and uptime alerts.

- Page loads under 2 seconds for normal use; the quote PDF is generated in under 10 seconds.

Usability

- Responsive web app, usable on a phone. Plain, fast interface; an agent should be able to submit a request in under 3 minutes.
- Accessible contrast and keyboard navigation for the main flows.

Maintainability

- Automated tests for pricing, margin, commission, invoice numbering and permissions (the places where bugs cost money).
- Documented code, setup instructions and an architecture diagram.

8. Operator qualification method and criteria

The company is the contracting party, so its exposure is greater than a pure agent's. Qualification should be documented, repeatable and checked on a schedule. Suggested method: **document check, third-party safety rating, then a conversation, then a conditional approval and periodic review.**

Mandatory (all must pass)

1. **Valid Air Operator Certificate** for commercial air transport, issued by a recognised authority, with operations specifications covering the aircraft type and the type of operation requested. Verify with the authority's records where possible.
2. **Aircraft on the AOC:** the specific registration is listed on the operator's AOC (no sub-charter without written permission).
3. **Valid airworthiness documents** for the specific aircraft, such as the Certificate of Airworthiness and Airworthiness Review Certificate.
4. **Insurance certificate:** aviation liability and hull cover with limits meeting or exceeding the legal minimum and the company's own minimum (the company sets this), covering the operation and territory, with the company named or an agreed indemnity in place. Check expiry.
5. **Independent safety audit rating** from a recognised provider. Options include ARGUS (Gold, Gold+ or Platinum), Wyvern (Wingman or Registered) and IS-BAO registration. The company chooses which rating levels are acceptable. Verify the current rating directly with the provider and the scheme's current tiers.
6. **Two-pilot crew** with licences and medical certificates in date, and minimum experience set by the company.
7. **24/7 operations control** contact.

8. **No sub-chartering** without written consent and re-qualification of the sub-operator.
9. **Signed operator agreement** with the company covering insurance, liability allocation, cancellation, substitution, change and operator-identity terms.

Scored or conditional

- Safety management system in place; incident and accident history (reviewed, with explanation required)
- Aircraft age, interior condition and maintenance programme
- Financial standing and payment history, including any refund incidents
- On-time performance and response time to quote requests (from the platform's own data)
- References from other brokers
- Fleet depth: ability to substitute an aircraft

Process

1. Operator submits or admin uploads documents; checklist ticks pass or fail with evidence.
2. Admin decision: Approved, Approved with conditions (for example, specific aircraft only, or specific regions), or Rejected.
3. **Re-check:** automatic expiry alerts, annual full review, and immediate review after any incident or complaint.
4. Scorecard updates after each trip. Low scores trigger review.

The company should also ask its aviation legal adviser and insurer whether any criteria are required for its own cover or regulatory position.

9. Integrations

Integration	Phase	Notes
Payment provider (Stripe or Mollie)	1	Payment links, webhooks, refunds
Transactional email (e.g. a mainstream provider with EU sending)	1	Delivery, bounces, templates, domain authentication (SPF, DKIM, DMARC)
Maps or geocoding	1	Distance and time use own calculations; avoid per-request mapping fees unless needed

Integration	Phase	Notes
Avinode	3	Phase 1 is manual. Avinode offers an API for brokers (leads, RFQ and quote download, search launch) that needs approval and a suitable plan. Design the sourcing model so imports can be added later without redesign
Other marketplaces	3	Manual logging in Phase 1
Flight tracking (ADS-B)	3	Check licence terms and costs
Accounting export	2	CSV first
Paid e-signature provider	Optional	Built-in signature in Phase 1

10. Phasing

Phase 1: working business (MVP). USR-1 to 6, AGT-1 to 7 and 10, ADM-1 to 8, QTE-1 to 9 and 12, CLI-1 to 7, DOC-1 to 7, PAY-1 to 4, REF-1 to 3, 5 and 6, OPR-1 to 6, NOT-1 to 3.

Phase 2: efficiency and control. Commission statements, operator payables, benchmarking, AI quote capture, scorecards, in-app notifications, trip changes, feedback, CO_2, country notes, accounting export.

Phase 3: scale. Avinode and other marketplace APIs, empty-leg watchlist, flight tracking, white-label agent branding, operator portal, WhatsApp and SMS.

Suggested milestones within Phase 1 (developer to confirm timing and fees):

1. Foundation: authentication, roles, audit log, reference data
2. Request and sourcing: agent portal, admin queue, sourcing log
3. Quote engine and quote documents
4. Client portal: acceptance, e-signature, passenger form
5. Invoices and payments
6. Flight brief and manifest
7. Operator qualification module, notifications, hardening, UAT

Each milestone is demonstrated on a staging site and signed off before the next payment.

11. Acceptance criteria for Phase 1

- An agent can register, submit a three-leg request and see its status; another agent cannot see it.
- An admin can log three operator responses, create a quote with margin and commission, and publish it; the agent sees the client price but **not** the operator name, operator price, margin or commission rules of other agents.
- The client opens the link on a phone, accepts, signs, enters two passengers, pays by card and by bank transfer (test mode); invoice status updates correctly in both cases.
- A quote revision creates version 2; the client sees only version 2; admins can see both.
- Invoice numbering has no gaps, even when two invoices are created simultaneously.
- A flight brief is generated from trip data, published, revised and re-published, and the client sees the version history.
- An operator with an expired insurance document cannot be selected in a quote.
- Audit log shows every price change and every view of passport data.
- Restoring from backup to a fresh environment succeeds.
- Automated tests cover pricing, commission, invoice numbering and permissions.

12. Open decisions for the client

ID	Decision	Impact
D-1	Can agents add their own markup, or is the client price always set by the company with commission paid separately?	Pricing model, AGT-9
D-2	Do quotes go to the agent only, or also directly to the client?	Notifications, portal
D-3	Operator-name disclosure point after legal advice	OPR-6, flight brief
D-4	Invoicing entity, VAT treatment and legal entity details (including the Malta structure, if it will invoice)	Invoice templates, DOC-3
D-5	Payment provider choice and card thresholds	PAY-1, PAY-5
D-6	Minimum insurance limits and acceptable safety-audit ratings	Section 8
D-7	Data retention periods for passenger identity data	Section 7
D-8	Agent commission structure and payout timing	QTE-3, AGT-8

ID	Decision	Impact
D-9	Cancellation and change policy for clients	Quote terms
D-10	Currencies to support beyond EUR	QTE-4

13. Out of scope for Phase 1

Operator portal, native mobile apps, automated Avinode integration, flight tracking, WhatsApp integration, accounting software integration beyond CSV export, crew and aircraft scheduling, and any flight-operations or dispatch functionality.